

A Distance-Dependent Model for Spatial Cell–Cell Interactions

Yalei Zhao, B.S., Dylan Cable, Ph.D.

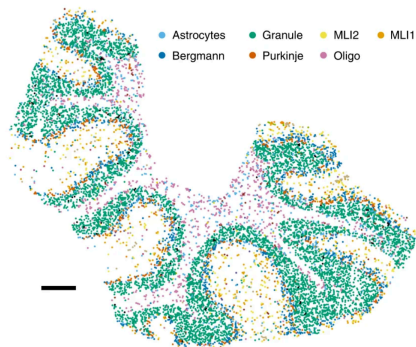
University of Michigan Department of Biostatistics

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Core problem



Does their spatial arrangement reflect meaningful interaction between cells, or just random co-occurrence?

- Why we should care? Cell–cell interactions are central to many biological processes (e.g., immune response, cancer progression, tissue development)
- **Importance of cell-level interaction:** provide the foundation for downstream ligand–receptor signaling and differential gene expression.
- Our goal is to quantifying interaction between cell types, and examine how that interaction changes with distance.

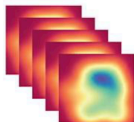
Existing approach - Spatial colocalization score

Colocalization score: quantify whether two cell types tend to occur in same spatial region.

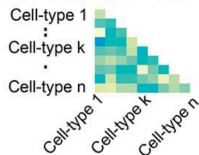
Example: Grisanti Canozo FJ, Zuo Z, Martin JF, Samee MAH

Workflow: spatial data $\rightarrow$ cell-type spatial kernel density $\rightarrow$ colocalization scores $\rightarrow$ filter candidate interactions $\rightarrow$ infer ligand-receptor signaling

Spatial density estimates



Co-localization scores



Limitations:

- Requires a predefined spatial region.
- Summarizes interactions and may lose detailed distance-dependent information.

Existing approach - Enrichment

Enrichment: compare observed neighboring cell type frequencies to a random expectation. **b**

Example: CRAWDAD by dos Santos Peixoto, R., Miller, B.F., Brusko, M.A. et al.

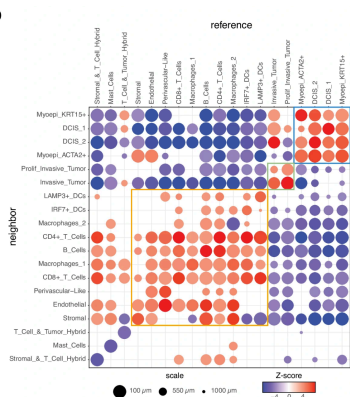
Workflow: spatial data → neighbor counts (within a fixed radius) → proportion of neighboring cell types → Z-score (enrichment/depletion) → infer interaction

Limitations:

- Depend on fixed radius
- Binary notion of interaction (enriched vs not)
- Cannot capture nonlinear interaction patterns

Our model:

- Models interaction as a continuous function of distance
- Quantifies interaction strength across spatial scales
- Captures nonlinear, distance-dependent effects



Cell interaction model structure

Idea: Model how the proportion of cell type A around each focal cell influences the probability that the focal cell is of type B, allowing the effect to vary smoothly with distance.

$$\text{logit } \Pr(B_i = 1) = \beta_0 + \int \beta(d) p_i(A \mid d) \, dd$$

(1) Local density via kernel smoothing

Estimate $p_i(A \mid d)$ using kernel-weighted counts of nearby cells.

(2) Flexible distance effect via splines

Represent $\beta(d)$ using spline basis functions.

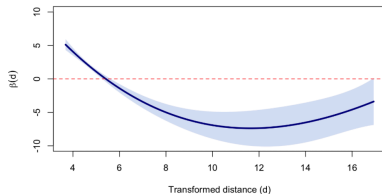
(3) Smoothness via regularization

Add a penalty on the second derivative of $\beta(d)$ and estimate the model using penalized maximum likelihood.

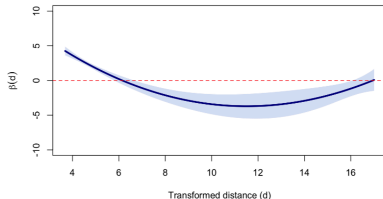
$$\max_{\beta} \ell(\beta) - \lambda \int (\beta''(d))^2 \, dd$$

Results: Purkinje–Bergmann

Cerebellum slice 08



Cerebellum slice 09

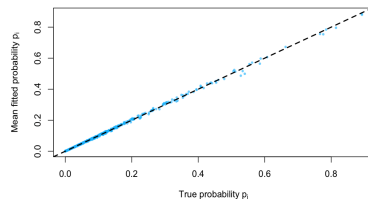


Distance-dependent interaction pattern is consistent across slices.

Model Comparison

Model	LogLikelihood
Baseline	-1554.1
Linear	-1530.9
Quadratic	-1495.4
Cubic	-1472.5
Penalized spline	-1397.2

Model Recovery Under Simulation



Conclusions

Modeling distance-dependent spatial structure provides additional information than summary statistics.

Interpreting spatial interaction

- $\beta(d) > 0$: presence of A at distance d increases the probability of B
- $\beta(d) < 0$: presence of A at distance d reduces the probability of B
- $\beta(d) = 0$: no distance-dependent effect

Biological insight

- Identify interaction ranges
- Complement downstream ligand–receptor analysis
 - Strong spatial cell interaction $\rightarrow$ increased likelihood of cell–cell signaling
 - Long-range effects $\rightarrow$ potential long-distance signaling

1. Tissue identification

- Different tissues show distinct interaction patterns
- Classify unknown tissue samples based on interaction patterns

2. Disease detection

- Learn interaction patterns in healthy tissue
- Compare new samples to the healthy reference to detect deviations in spatial organization

3. Potential medical applications

- Predict which cell types interact with a given cell type
- Identify effective interaction distances
- Inform strategies to modulate or block signaling pathways at specific regions

References

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Characterizing cell-type spatial relationships across length scales in spatially resolved omics data.

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